



Beyond Illusion

BUILDING ETHICAL DEFENSES AGAINST DEEPFAKES

| T7_FIT1055RRAH |

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Problem Statement

1. Real Case Studies

- (a) The app DeepNude proves that women's bodies can be violated with the misuse of technology. With just a single click, the app takes pictures of clothed women and produces shockingly realistic nudes, going as far as to replace clothing with genitalia and exposed breasts. While the free version of the app does add a watermark, anyone with malicious intentions and \$50 to spare can easily overcome this so-called hurdle, as the paid version produces images that are nearly impossible to classify as fake or real (Cole & Cole, 2024).
- (b) In Hong Kong, a worker was deceived into transferring \$25.6 million after falling victim to a video call where every other participant, including the company's CFO, was a deepfake. A massive reason as to why hackers were able to get away with this was due to how genuinely the "colleagues" looked and sounded. Deepfakes affect not only individuals but also global corporations (Chen & Magramo, 2024).
- (c) During Russia's invasion of Ukraine, a deep fake of President Zelensky asking for surrender was quick to spread across social media. Though quickly debunked, it's noteworthy how deepfakes during war can act as psychological weapons, eroding trust and spreading panic in high-stakes moments where truth is most critical (Wakefield, 2022).

2. Problem Statement

Deepfakes are inherently about the exploitation of the identity of people against their will and using their faces or voices as a means of deception. Beyond harming individuals, deepfakes "erode digital authenticity and undermine the trust that societies place in visual and audio evidence" by blurring the line between real and fake (Furizal et al., 2025). Trust is the foundation of society. When deepfakes destroy it, people stop believing anything they see or hear. The ethical harm also extends to dignity, as victims are objectified and manipulated. According to

Chesney and Citron (2018), "the creation of non-consensual deepfakes strips victims of dignity and autonomy, treating their identity as raw material for exploitation". These violations are widespread, affecting every part of life. This makes deepfakes an urgent ethical problem.

3. *Who Gets Hurt?*

Deepfakes are a rapidly developing technology that has affected a wide range of individuals. Those affected include celebrities, women, politicians, journalists, businesses, corporate executives, human rights activists, educators, students, influencers, legal professionals, and everyday internet users across all sectors (Wahab, 2025).

A large-scale study showed that 27–50% of respondents in the education sector were unable to distinguish between authentic and deepfake videos (Doss et al., 2023). As stated in the briefing by the European Parliament, in 2024, 49 % of companies experienced audio and video deepfakes.

New research shows that, in 2024, the use of generative AI-based deepfakes increased by 118 % and that 90 % of US companies experienced cyber fraud. They further highlighted that children's cognitive abilities are still developing, and they face more difficulty identifying deepfakes, due to which they are more susceptible to harmful online practices, including grooming, cyberbullying and child sexual abuse material.

The key findings of a case study by Security Hero conducted in 2023 revealed that: 99% of the individuals targeted in deepfake pornography are women. 94% of those featured in deepfake pornography videos work in the entertainment industry. 74% of users don't feel guilty about it (*2023 State of Deepfakes: Realities, Threats, and Impact*, n.d.). This was data two years ago, and now the numbers have only multiplied over two-fold, as mentioned by the European Parliament. In recent years, deepfakes have doubled in number every six months. A projected 8 million deepfakes will be shared in 2025.

Therefore, this growing accessibility of deepfake tools commands the urgent need for counter-technology to help build a safer digital community, which is the main purpose of our solution.

4. *Why Current Solutions Fail*

Even though there is an increasing awareness of the evils of deepfakes, the existing measures are still very limited and ineffective.

One of them is the detection lag: the improving generative models are always ahead of detectors, which makes most systems outdated and incapable of reliably detecting new forms of manipulated content (Alrashoud, 2025) (Balafrej & Dahmane, 2024).

The activity of platform moderation is also mainly reactive; the harmful content is often removed only after the virality, and by that time, reputational and psychological harm caused to the victims is irreversible (Birrer & Just, 2024).

Laws have high levels of gaps and inconsistencies. Although there are cases of jurisdictions enacting laws that concentrate on political misinformation, non-consensual sexual deepfakes have little protection, and most victims, especially women, lack a proper solution to this (Guerrero-Sierra et al., 2025) (Mania, 2022).

Technical flaws also undermine existing solutions: detectors are subject to false positives, false negatives, do not work well on compressed social media data, and are inaccessible to non-specialists, e.g., journalists and victims (Balafrej & Dahmane, 2024).

Lastly, despite the significance of media literacy campaigns, they cannot keep up with the pace of technological change; that is, common users can hardly realistically discriminate between genuine and fake media (Birrer & Just, 2024).

Consequently, bad materials are continuously spreading, and the popularisation of digital media is gradually being undermined.

ERF Analysis Integration

Step 1: ACM Code Analysis

Deepfake is a severe violation of several concepts in the ACM Code of Ethics and Professional Conduct. Deepfakes are frequently misapplied instead of being employed in beneficial purposes, particularly on the female population and vulnerable populations.

→ Principal 1.2 Avoid Harm

Deepfakes cause major reputational, psychological, and financial damage. Victims may lose jobs, suffer harassment, or face long-term trauma. Non-consensual pornographic deepfakes are especially harmful: research shows that 95–96% of all deepfake videos are pornographic and almost all target women (Romano, 2019). The ACM Code stresses that computing professionals must work to reduce unjust consequences. Our detector will be used to avoid these harms by identifying and marking fake content at an early stage, before it becomes widely spread. In such a manner, it is possible to minimize reputational and psychological harm, and the victims can have an opportunity to defend themselves through evidence.

→ Principal 1.3 Be Honest & Trustworthy

The entire purpose of a deepfake is to misrepresent reality, trick viewers, and erode trust in media. The ACM Code requires honesty and integrity because “honesty is an essential component of trustworthiness” (*The Code Affirms an Obligation of Computing Professionals to Use Their Skills for the Benefit of Society.*, n.d.). When computing professionals release systems that are designed to deceive, they directly undermine this value. Our detector supports honesty and transparency by clearly showing when content has been manipulated. Instead of

leaving people uncertain, it provides a risk score or label, restoring trust by helping people distinguish between real and fake media.

→ Principal 1.1 Contribute to Society and Human Well-Being

AI technologies should be built to benefit society, yet deepfakes show the opposite. When AI tools are used to humiliate victims, spread misinformation, or create fraud, they fail to reach this standard. Our detector helps restore this principle by giving individuals and organizations tools to identify misuse. Victims regain dignity, platforms reduce harmful content, and society benefits from a safer digital environment. In this way, the detector supports well-being instead of harm.

→ Principal 1.6 Respect Privacy

A person's face, body, or voice is taken without consent and reused in harmful contexts. ACM states that "personal information gathered for a specific purpose should not be used for other purposes without consent" (*The Code Affirms an Obligation of Computing Professionals to Use Their Skills for the Benefit of Society.*, n.d.). Deepfake creators ignore this completely. Our solution will respect privacy by providing users with control: in case their likeness is abused, they will be able to request takedowns, provide evidence, and challenge platforms. This replenishes a certain power of victims and safeguards the right to personal image.

→ Principle 2.5 Comprehensive Evaluations

Another major issue is the lack of safeguards before releasing deepfake tools. Extraordinary care should be taken to identify risks in computing systems, especially in AI and machine learning. However, many developers release open source deepfake tools without risk assessments, testing, or review. They prioritize technical achievement over responsibility. Our detector does the opposite: it includes built-in accuracy checks, reliability scores, and third-party audits before launch. This ensures misuse risks are considered and reduced.

→ 3.1 Public Good as Central Concern & 3.7 Recognize and Mitigate Risks

Developers and platforms must put the public good above private profit and must recognize and address risks. Ideally, platforms tend to give space to malicious deepfakes as viral content attracts more clicks, engagement, and revenue. Misuse of their models is also hardly anticipated by developers who concentrate on the speedy release. This puts society at risk. Our detector addresses this by putting safety in the first place as compared to virality. It also involves misuse scenario testing and threat modelling before the launch, and risks are not overlooked.

Overall, deepfakes are unethical in accordance with several principles of ACM Code: they are harmful, misleading, support discrimination, disrespect privacy, and are not properly evaluated, as well as prioritizing profit over the good of the people. Our deepfake detector is created as an ethical reaction. It can identify malicious material in advance, enhance transparency, emphasize equity, guard consent, and put safety measures into its foundation. Our design is in accordance with the principles of ACM to demonstrate that the use of AI can be responsible for keeping society safe, instead of endangering it.

Step 2: Stakeholder Impact

Many groups are impacted by deepfake technologies, yet four of them have the highest risks and are, therefore, the most beneficiaries of our proposed detector: victims, organizations/businesses, media and journalists, and society in general.

1. Victims

The most vulnerable stakeholders are the victims, particularly the women. Research indicates that 95–96 per cent of deepfakes are pornographic and disproportionately target women against their consent (Romano, 2019). The ease of exploitation can be seen with such apps as DeepNude, which can generate non-consensual sexual images of any woman with the press of a button (Cole & Cole, 2024). Such abuses are against dignity, privacy, and safety, and

result in trauma and rupture of reputations among victims (Guerrero-Sierra et al., 2025) (Mania, 2022).

Our detector provides a victim portal where anyone can post suspected fake content. It will confirm whether their likeness is being used improperly and automatically create tamper-proof bundles of evidence (with timestamps and digital hashes) that may be presented in a takedown request or in court. This offers quicker relief and gives victims practical tools to reclaim authority.

2. Organizations & Businesses

Deepfakes are starting to be used to commit fraud and impersonation. In a recent example in Hong Kong, a finance employee lost 25 million dollars after being duped by a deepfake video call of their CFO (Chen & Magramo, 2024). Businesses are thus premium targets who risk losing finances, reputation, and trust in virtual communications.

Our detector offers an authenticity verification layer for corporate communications. The system verifies whether a video or audio feed is authentic before sensitive decisions are made. In this way, it avoids fraud, protects organizational property, and helps maintain trust in digital cooperation. This is critical as deepfake frauds are becoming more sophisticated and larger in scale (Alrashoud, 2025).

3. Media & Journalists

Journalists and media are exposed to reputational and professional dangers when tampered content is posted as fact. In the Russia vs. Ukraine war, a viral video of President Zelensky calling on his troops to surrender was faked, threatening credibility during a conflict (Wakefield, 2022). Media professionals need the means to authenticate content prior to release.

Our solution includes a newsroom plug-in integrated into editorial processes. Journalists can run suspect media on the detector, receive risk indicators, and view heatmaps of manipulated areas. This minimizes liability and safeguards

credibility. Restoring confidence in information systems requires not only promoting literacy but also embedding technical protection measures in workflows (Birrer & Just, 2024).

4. Society at Large

Lastly, society is at risk of being affected collectively by disinformation through deepfakes. Studies point out the endangerment of democracy, privacy, and national security because deepfakes have already been applied in political manipulation, harassment, and propaganda. The mistrust in the online content leads to confusion among the voters and poses a hindrance to the democratic process.

Our detector helps us to increase resilience by avoiding misinformation even before it can go viral. It adds more value to the information ecosystem through offering users' authenticity cues, platform-level moderation and forensic support to regulators. The result is regaining faith in digital media and enhanced democratic procedures. The European Parliament claims that minors and youths are particularly vulnerable to manipulation, and therefore, measures should be in place to guarantee a healthier digital future.

The most acutely affected groups of deepfakes are victims, organizations, journalists, and society. Each suffers harms such as sexual exploitation, financial fraud, and destabilization of democracy. These harms are directly tackled by our detectors through proactive detection, evidence generation, authenticity verification, and scalable moderation tools. By so doing, it complies with the ACM principle of putting the public good first while ensuring dignity and security for those most at risk.

Step 3: Solution Justification

In the deep fake detectors as discussed in the articles by SOCRadar and StartUS, there are numerous high-performing and effective detectors in the deepfake detection

market, but our multi-platform real-time detector has a different take on the market that includes filling the gap between consumer usability and enterprise-level performance.

Most of the solutions available are designed to support large organizations, whereas our system has been designed to support both everyday internet users and businesses. It is compatible with browsers, desktops, mobiles, and work points through the usage of apps, extensions, and integrations. It is multi-modal, supporting manual uploads of images, audio, and video in one free package.

SOCRadar highlights the features and flaws of several detectors. For example, OpenAI's DALL-E 3 achieves high accuracy on its own images but only 5–10% on others. Hive AI labels each face as “yes_deepfake” or “no_deepfake” with a confidence score, a feature we extend with clear explanations (e.g., lip-sync or blinking inconsistencies). Intel's FakeCatcher, launched in 2022, analyzes biological signals via photoplethysmography (PPG) to identify authenticity within milliseconds, achieving 96% accuracy. However, it focuses only on video, not audio. Our solution expands real-time detection across video, audio, and images.

Sensity AI provides real-time monitoring for governments and enterprises, analyzing multiple media types, with accuracy rates of 95–98%. It is accessible via a web app, API, and SDK, but lacks a ready-made browser extension. While they integrate with Microsoft Teams, we plan broader collaborations with multiple conferencing platforms. Importantly, both Sensity AI and Intel's FakeCatcher lack a free version for general users (Yao, 2022). By contrast, our model prioritizes accessibility for everyday users as well as enterprises.

Balancing Competing Interests

→ Free Speech vs. Harm Reduction

Enterprise detectors, most of them, are binary decisions, which can lead to censorship. The classifier of OpenAI merely identifies AI-generated text, whereas Pindrop Security has 99 per cent accuracy, which is only applicable in call

centers, not in consumers (Owda, 2025). As deepfake attacks are estimated to hit 9.9 billion in 2027 and have a 5 billion dollar income (Burt, 2025), the demand is growing. Our style of explainable confidence prevents the issue of censorship by displaying users' reasons why the content is fake, just like the McAfee extension that indicates the presence of manipulated audio pieces (McFarland, 2025). This gives the users the power, without silencing the content.

→ Speed vs. Accuracy

Across the market, there is a trade-off. Intel's FakeCatcher maintains 96% accuracy under controlled conditions but drops to ~91% on "wild" videos, where lighting and resolution interfere. The audio detector provided by Surf Security is quick (less than two seconds); however, it lacks analysis depth. Even with the Department of Defense support, Hive AI acknowledges that it has no unbeatable detector, and enemies evolve (Owda, 2025). Our tiered architecture is a solution to this: a lightweight, real-time scan marks suspicious content as such, and then forensic-granite analysis is applied to higher-risk cases. This guarantees quick warnings and in-depth accuracy, which few consumers tools offer.

→ Privacy vs. Transparency

Enterprise tools usually use cloud processing, which poses a threat to data security. By not storing audio or browsing history, McAfee demonstrates that it is possible to preserve privacy on detection (McFarland, 2025). However, YHAI, Reality Defender and Attestiv rely on off-server or server managed by an enterprise and need user trust. Even open-source FaceForensics++ will require datasets and knowledge (Owda, 2025). We design our systems to be local-first so that sensitive media does not leave the machine, hence maintaining privacy and providing transparency through cryptographic verification and explainable output. In contrast to cloud-only models, we combine privacy and accountability, which enhances the trust of the user.

Our detector deals with the fundamental flaws of the existing solutions by offering enterprise-level precision with consumer-level access. It does not require trade-offs between competing values, such as free speech and harm reduction, speed vs accuracy, and privacy vs transparency. This puts it in a unique position to serve individuals and organizations in a digital space that is becoming more threatened by deepfakes.

Product Solution

Product Overview

Our product is a multi-platform and real-time deepfake detector that allows ordinary individuals and organizations to recognize manipulated videos, audio and images with the accuracy of a forensic tool, without any loss of privacy or legal evidence to take down or prosecute offenders.

❖ Key – Features

The current deepfake detection landscape, as highlighted by SOCRadar’s *Top 10 AI Deepfake Detection Tools to Combat Digital Deception in 2025*, shows impressive technological progress but also clear fragmentation.

Many solutions are either enterprise-only, restricted to a single media type, single platform or operate after damage has already been done rather than real-time. Our solution bridges this gap by combining consumer accessibility, enterprise adaptability, and forensic-grade rigor into a single package. The features outlined below illustrate how we complete the ecosystem instead of competing directly with existing tools.

1. Real-Time Detection in Browser

A core differentiator is detected directly in the browser, in real time. Upload-based detectors require files to be sent to the cloud, then waiting hours for results and not to forget, privacy risks. By analyzing content at the point of consumption, our system prevents harm before manipulated media spreads. Intel’s

Newsroom article about the launch of FakeCatcher highlights the importance of real-time analysis. We democratize this by embedding detection into everyday browsing, transforming deepfake detection from an afterthought into a continuous safety net.

2. Probabilistic Scoring with Explanation Cues

Most detectors still give binary outcomes as “real” or “fake”. This approach often creates more problems than it solves, as false positives can damage trust and false negatives can allow harmful content through. Our system provides probabilistic scores (e.g., “70% fake”) and highlights explanation cues like lip-sync mismatches or unnatural blinking. This boosts digital literacy, builds trust, and reduces over-censorship. It mirrors Intel’s use of photoplethysmography (PPG) signals but makes the reasoning transparent to non- technical users.

3. Multi-Platform Protection

Where many competitors limit their scope, our tool works as a browser extension, desktop app, mobile app, workplace integration, and even video-conferencing platforms, mirroring Grammarly’s cross-platform presence. For organizations, we also provide APIs and SDKs to embed detection into workflows like video calls, content moderation, or fraud prevention and customize it according to their profession, enabling them to keep or remove any features we’ve provided. Unlike enterprise-heavy solutions such as Reality Defender or Hive AI, ours ensures ordinary users are equally protected.

4. Multi-Modal Detection

Deepfakes extend beyond video to include corporate audio scams, image-based misinformation, and fake identity documents. Our solution detects video, audio, and images, offering full-spectrum protection. Companies like McAfee and Surf Security specialize in audio only, while Intel’s FakeCatcher focuses on video

as covered in the product demonstration video by Dr. Ilke Demir. We cover every medium in one system.

5. Manual Upload Support

In addition to real-time browsing, users can upload suspicious media be it video, audio, or images for verification. This supports journalists, researchers, and impersonation victims who need to analyze specific files. While Attestiv offers this for enterprises with API integration, with a free version of 5 video uploads per month, as mentioned in SOCRadar’s article, our system makes forensic-grade uploads free and accessible to all.

6. Photoplethysmography (PPG) Integration

“When our hearts pump blood, our veins change color. These blood flow signals are collected from all over the face and algorithms translate these signals into spatiotemporal maps. Then, using deep learning, we can instantly detect whether a video is real or fake”. This is a detailed explanation of the procedure implemented in PPG Integration as explained in an article by Intel’s newsroom. Since this method is proven to have an “accuracy of 96% returning results in milliseconds” in real-time, this feature would further enhance the functionality of our solution.

7. Privacy-Preserving Architecture

Privacy is one of the biggest concerns in deepfake detection. Many tools depend on cloud-based processing, which forces users to trust that their sensitive media won’t be misused or retained. McAfee, for instance, states that it does not store user audio or browsing history, and it highlights on-device processing using Intel hardware. However, the specifics of how data is handled remain largely undisclosed to users, leaving the process less transparent.

Our approach removes this uncertainty. By adopting a local-first architecture, all detection happens directly on the user’s device. Nothing is uploaded or retained, and no browsing or audio data ever leaves the system. This not only aligns with

global standards for privacy-preserving AI but also gives users confidence that their personal information remains under their control.

8. Evidence Export for Action

Detection alone is insufficient for victims, investigators and even journalists. Our tool can generate evidence bundles upon prompting, that document scores, cues, and metadata in standardized formats. This supports reporting, takedowns, and legal action, going beyond consumer alerts offered by McAfee or Surf.

9. Enterprise API Integration

While designed for everyday users, our system also scales to organizational needs which will be mentioned in more detail below. Enterprises and platforms can embed detection into moderation and security workflows via APIs. Unlike Reality Defender or Hive AI, our platform is not locked behind enterprise paywalls but functions as a complete ecosystem serving both consumers and professionals.

❖ Target Users

The harms of deepfake technologies are ubiquitous, but certain groups are at a disproportionately high risk of harm and thus will benefit the most as a result of our proposed real-time, multi-platform detector. Such victims are those of non-consenting material, the ordinary internet users, journalism and media houses, agencies and businesses, social networking sites, and law enforcers or juridical practitioners.

1. Victims of Deepfakes

The most susceptible users are the direct victims, especially women. The effects of such abuses include reputation damage, emotional distress, and, in most cases, no productive path to justice (Guerrero-Sierra et al., 2025) (Mania, 2022). In this case, our detector gives a victim portal where users can upload suspicious deepfake content. The system ascertains the possibility of their

likeness being abused and creates tamper-proof evidence packages with timestamps and digital hashes. These bundles may be provided in takedown requests or in court testimony, providing an acknowledgement of redressability and a useful means of redress. This design is empowering to the victims and puts the most vulnerable people on the frontline in terms of their dignity and safety.

2. General Users

Protection is also needed by everyday users of the internet. With the creation of viral deepfakes, there is much confusion, as it destroys the self-esteem of digital information. The general user is not knowledgeable enough to know the genuine and the fake, and they fall prey to being duped. Our detector is built into browsers, mobile phones, and computers, giving out authenticity clues and risk ratings prior to the exchange of media or trust in the media by users. Such transparency will enable individuals to make their decisions independently without the limitation of the platforms or third-party censors. According to Birrer et al. (2024), developing trust in information ecosystems needs media literacy and technical backups, which our system provides in an easy-to-use manner.

3. Reporters and Media Outlets

In publishing manipulated content, media professionals would be at reputational and professional risk. The case of Zelensky also showed that fake media can damage the credibility of reporting on critical occasions. False reporting is harmful not only to the credibility of individual outlets but also to the press (Chesney and Citron, 2018). In this regard, our detector is a newsroom workflow that can be integrated with editorial workflows. The system allows journalists to pass suspect content through the system, clear risk scores, and heatmap explanations of parts of the image that were manipulated. This minimizes liability, credibility is safeguarded, and authentic media is only

published. This tool is directly helpful in the task that journalists have to deliver quality information in a digital environment full of manipulations.

4. Organizations and Enterprises

Financial deepfake scams have already posed as victims of enterprises. This kind of fraud poses threats to the organizational budget, and it undermines the credibility of online communication platforms. Our detector minimizes these threats by offering authenticity verification layers on corporate communications. The businesses can also check the authenticity of a video or audio interaction before any decision is made, or transfers are made. This fraud-detection tool helps secure the assets of the corporation, provides secure communication, and gives the partners and employees confidence. The pace of detection method development should keep up with the ever-more-advanced attacks, and the integration of various enterprises is necessary (Alrashoud, 2025).

5. Social Media Platforms

Much of the harmful content is hosted by platforms, and they are becoming increasingly accountable. However, the moderation that is presently enforced is mostly reactive, whereby content is only removed after it has become viral. This puts the users at risk, as well as tarnishing the reputation of platforms. We offer an automated moderation pipeline API so that platforms can flag or block suspicious uploads in real time. This minimizes damage, enhances user safety and decreases the legal and reputational risk of platforms. Detection during the upload stage will make moderation less reactive than proactive, which is more reliable to the safety expectations of the population.

6. Law Enforcement and Law Authorities

Lastly, the police forces and the judicial system need mechanisms that could be used to convict criminals. According to Mania, 2022 and the European Parliament, the existing legislation cannot keep up with the misuse of deepfakes

in sexual exploitation and fraud. Investigators must have forensic-grade evidence that can withstand challenges. We are responding to this in our detector by generating legal evidence bundles that cannot be tampered with. These bundles have good support in lawsuits with the help of timestamps, cryptographic hashing and transparent detection logs. This allows prosecutors and regulators to work more efficiently in chasing the offenders, and the victims have more powerful cases before the court.

❖ Our Edge

The deep fake detection industry has developed at a high pace, with several firms introducing tools to overcome the evils of the manipulated media. Nonetheless, the present solutions are still of a narrow scope, availability, or flexibility. Image-only detectors are found, and they have a narrow concentration on one type of media. Others are reactive, which means that they can only be implemented when damage has been done, or are limited to enterprise and government use, with mere users not protected.

There are also regional or language-specific tools that are inclusive to a small subset of the world population. It is in this background that our detector comes with a number of advantages that will distinguish it.

1. Consumer-Friendly Accessibility

Most of the available solutions are designed to focus on corporate or government customers, thus leaving individuals (who are often vulnerable to harm) without any effective tools. As an example, Sensity AI mostly targets business clients, cybersecurity organizations, and states, and its APIs and SDKs are not user-friendly without technical skills. In our case, however, our detector is universal. It operates using browser extensions that are easy to use and mobile applications and desktop integrations that enable professional-level detection to be accessible to normal internet users. Such democratization of access has the benefit of enabling protection to benefit

not only the victims, but students, parents, and journalists alike, without requiring enterprise resources.

2. Real-Time Protection

Reactive moderation is still a significant weakness in the industry. On social media, such as Facebook and Twitter, it is common to see fake information spread before being removed, as it has become viral, and the damage to reputation and psyche is already inflicted. Although the FakeCatcher, introduced by Intel, was sold as the first real-time deepfake detector, it is limited to the analysis of videos only, and it does not apply to audio or images (Hutson, 2023). Our detector will provide multi-modal and real-time scanning of suspicious media, whereby a flag is raised on video, audio, and picture files immediately. This will ensure that unhealthy content does not spread prior to reaching mass audiences, and the paradigm changes to proactive prevention rather than damage control.

3. Privacy-Preserving

Design The other issue with most enterprise applications is that they use cloud computing, which increases vulnerabilities to data leakages. As an illustration, Hive AI and Reality Defender rely on off-the-shelf servers, where users must rely on third parties to manage sensitive uploads (Owda, 2025). The solution is based on a local first architecture, meaning we work without storing uploads and make sure that they are done on the gadget. Cryptographic validation and tamper-proof logs are transparent, although not invasive of privacy. This trade-off provides users with an assurance that no one will abuse their personal media, yet it gets good detection results.

4. Multi-Lingual and Localizable

Another weakness of the existing systems is language and territorial constraints. Most of the detectors are English or Western optimized and less effective in other cultural and linguistic environments. Our detector is

multi-lingual in nature, being able to adjust to various scripts, voices and local video standards. Such cross-cultural flexibility is not just a business product targeted at the Western markets but a tool that can help the people of the world, such as revenge porn victims in Asia or citizens facing political deepfakes in Europe and Africa.

5. Explainable Probabilistic Results

Numerous tools now give binary results, including Microsoft Video Authenticator, whose only result is to indicate whether an image is probably real or fake. Binary labels have the potential for false positives (censorship) and false negatives (no harmful content identified). Our detector is probabilistic with scores of confidences and explains what went wrong, including the presence of lip-sync errors or unnaturally blinking. This will enable users to make subtle decisions rather than being restricted to the opaque yes/no categories. Like in the explainable audio detector of McAfee (McFarland, 2025), our model will focus on transparency and empowerment of users.

Product Impact

❖ Immediate Benefits

→ Empowering Victims and Protecting Personal Identity

Our tool will provide rapid support to the victims of non-consent deepfakes by providing a dedicated portal to upload the suspected content and verify the misuse and produce evidence packages that cannot be tampered with (timestamps, hashes). Such packages assist in takedowns and lawsuits, as they offer victims an effective means to reassert their online identity. This makes the process easy and reachable and when the victims do it, they are able to take fast action, reduce the time of damage and recover both dignity and peace of mind.

→ Enhancing Journalistic Integrity

In the case of journalists, our detector will guarantee increased accuracy and confidence in reporting. It offers probabilistic authenticity scores as well as explanatory cues which would assist the newsrooms to check out a piece of information prior to publication. This not only shields outlets against any reputational damage, but also builds trust among the people in journalism, as it makes sure that citizens get quality information. The tool provides a strong protection of the truth and accountability in democratic society, because it advantages journalists by lessening the dividend of the liar, who fails to recognize authentic content as credible information.

→ Protecting Companies against Fraud and Impersonation

Proactive fraud prevention is beneficial to business. Our detector provides an authenticity validation layer to corporate communications, such as video calls, voice messages and sensitive files. This makes it so that fraud cases or a case of impersonation can be detected before it is too late. Organizations are thus able to secure their assets, their brand and retain stakeholder confidence. The detector does not only thwart expensive fraud, but it also provides long-lasting resiliency to the enterprise communication systems.

→ Assisting Platforms in Content Moderation

The social media sites get a moderation solution that is scalable. Our solution can be configured to be seamlessly integrated through real-time APIs, which automatically tag, label, and rank suspicious uploads so that they can be reviewed. This allows platforms to preempt viral harmful deepfakes, minimizing reputational risk and making online spaces safer.

Proactive detection during the uploading phase allows platforms to have pipelines that are more efficient and faster with moderate speed and better human reviews, which enhances user safety.

→ Facilitating Legal and Forensic Investigations

Our detector provides forensic quality evidence to law enforcement and courts. The system produces detailed database reports with metadata, manipulation heat maps and digital signatures that can be produced in the court. This makes investigations and prosecutions stronger, and it is difficult to escape the justice. The fact that manipulation can be clearly and verifiably proven is an advantage to legal authorities who can much more effectively control the enforcement of digital rights and offer victims a better avenue to justice.

→ Promoting Digital Literacy and Public Awareness

The contribution of our detector to long-term resilience is that it increases digital literacy. The system has explanation cues and risk scores that are immediately available in the interface, giving users the reason behind the content being considered manipulated. This makes it a learning experience where online media will be critically assessed by people, and the skillset is developed. According to the research, the people can become more effective at identifying deepfakes with the help of digital literacy. When we incorporate these explanations into the tool, we will be making these users more confident, resistant to manipulation, and capable of making sound decisions on the internet.

❖ Broader Implications

→ Rebuilding Trust in Online Ecosystems

Deepfakes have deep-rooted effects on society, and their consequences on individuals are merely the tip of the iceberg. According to

Pawelec (2022), exposure to deepfakes causes the erosion of trust in society as it weakens faith in the media, institutions, and citizens. Such a loss of common truth poses a danger to the very principles of democracy, where only informed debate can be based on a reality principle. Our product restores trust in digital ecosystems by providing real-time authenticity checks and explanation cues. By so doing, it enhances communicative features, eliminating confusion and restoring confidence in real media.

→ Defending Democratic Processes

Deepfakes are becoming more politicized, and fake videos of politicians are used to undermine elections or international relations (Wakefield, 2022). According to the European Parliament, young people are particularly susceptible to the manipulative media. Our multilingual scaling detector can be incorporated into election monitoring machines to identify dangerous content before it can affect the voter. This tool will make the democratic discourse safe because it will minimize political misinformation and make sure that citizens base their choices on reality, rather than falsehoods.

→ Endorsing AI Ethical Adoption

The AI technologies are viewed with doubt in cases of misuse to manipulate. In comparison, responsible innovation is exhibited by means of such tools as ours. Privacy-aware design, consent check modules and explainability aspects demonstrate that AI can be made in a manner that does not violate human dignity and freedom. This will not only appease users but also act as a template for other AI governance structures in the future. The detector integrates the elements of ethics in the product design, furthering the ACM requirement that means serving the public good, rather than profit (*The Code Affirms an Obligation of Computing Professionals to Use Their Skills for the Benefit of Society.*, n.d.).

→ Minimizing Harassment and Enhancing Mental Health

Deepfakes are devastating psychologically, particularly to victims of sexual exploitation. The content that occurs without consent may lead to permanent trauma, anxiety, and isolation (Guerrero-Sierra et al., 2025) (Mania, 2022). Our product allows making takedowns fast and checking consent, thereby minimizing exposure, regaining control, and authenticating victims. This not only reduces short-term suffering but also portends that some cultures consider such evils to be unacceptable. A more secure online space is a factor that has a direct positive impact on personal and community well-being.

→ International and Policy Implications

Deepfakes are cross-border and pose a threat to global security. As an example, fake news about political leaders can be used as a weapon to initiate a crisis. To deal with such threats, it is necessary to engage in international cooperation and moderate regulation. The use of detection tools can be used in a responsible way and help governments to defend elections and diplomacy without violating privacy and free speech. Our product is the perfect example of such a balance with powerful protection, but without censorship. In such a way, it contributes to the creation of new legislation and international regulations regarding AI, privacy and cybersecurity.

→ Economic and Technological Implications

There is a rapid change in technology due to the war of arms between deepfake makers and detectors. Though the fake will keep on being more convincing on the part of the adversaries, our detector will keep it in check by retraining continuously and updating the models. This flexibility means that it adapts to the threats. The market is likely to grow

exponentially in demand for deepfake detection and is expected to be in the billions of dollars in the next few years (Burt, 2025). Our solution, offering a convenient user-friendly product, will not only remove harm but also project itself as a sustainable and ethical participant in an expanding industry.

❖ Scalability Potential

→ Expansion to multiple languages

One of the biggest challenges faced in the expansion of deepfake detection is language diversity. While the world communicates in thousands of languages, each language having its own sounds and rhythms, existing systems are trained mainly on English data sets (Hou et al., 2024).

A detector that works well in English videos would not be able to detect deepfakes in other languages like Mandarin, Hindi, Arabic, etc., as lip movements, accents, and audio signals differ from language to language. To overcome this, we require large multilingual data sets and retraining of models to handle different phonetic structures so we can ensure that the performance of our model doesn't degrade when faced with a deep fake that isn't in English.

→ Collaboration with telecom and caller id services

Deepfake scans target organizations through real-time communication. One example worth noting is when WPP CEO Mark Read was targeted in an elaborate scam where fraudsters set up an AI-generated voice and manipulated video footage during a Microsoft Teams meeting with the goal of impersonating senior executives, including a fake WhatsApp account and footage taken from YouTube, to extort personal details and money (Thomas, 2024).

If we want our deepfake detection software to be truly scalable, it will have to connect directly to caller ID services that will verify in real time if the

voice/video on the line is genuine or manipulated. Combining AI detection alongside the existing Telecom authentication methods could provide an early warning system for scam calls. Preventing losses before they happen rather than resorting to analyzing evidence afterwards (Alrashoud, 2025).

→ Integration into legal and government sectors

For large-scale deployment to work, our deepfake detection must align with the legal standards. The “EU AI ACT” insisted on the disclosure of generated or manipulated content.

This is one of many such examples, and a scalable detection system should be designed to fit within these frameworks. By aligning our software to these laws and regulations from the start, it will be much easier to deploy across multiple industries, making it seen as a tool that supports both individuals and industries (*EU Artificial Intelligence Act | Up-to-date Developments and Analyses of the EU AI Act*, n.d.).

→ Monthly retraining

With how quickly deepfake technology is evolving, static models quickly become obsolete. Our detection system is committed to a continuous retraining system that ensures every update will include samples of emerging deepfake techniques and evaluation across multiple platforms (Birrer & Just, 2024). It is proven that models perform well in one data set but don't perform very well when put up against new manipulations. Regular training ensures that our system will catch new threats efficiently, sending users a message of trust and accuracy (Balafrej & Dahmane, 2024).

→ Collaborating with professionals for fact checking

Automation software will not always be able to capture intent or context. Independent verification by people adds essential human judgment. Fact checkers provide us with vital independent verification and will

publish corrections that build public trust. Their feedback can also be used to improve future versions of the models. By incorporating people, our software isn't just a detector anymore; it has become a part of a larger ecosystem that includes general society, journalists, tech professionals, etc.

Conclusion

Deepfake technology is one of the urgent ethical and social issues nowadays. Since non-consensual sexual exploitation to political disinformation, corporate fraud, etc., the damage it causes takes away individual dignity, institutional trust, and democratic stability. Our project has demonstrated that these harms are not accidental but are based on the breach of major ethical principles especially those in the ACM Code of Ethics which includes the requirement to avoid harm, respect privacy, and prioritize the good of the populace over personal gain. Any current solutions are helpful, but have been found to be too restrictive, too response based or too selective to offer sufficient safeguarding to the most vulnerable members of society.

The Ethical Reasoning Framework (ERF) analysis showed that there are numerous stakeholders who are victims of deepfake abuse. Women in particular and the victims, in general, are disproportionately harassed and their reputations are damaged. In a world where truth is readily fabricated, journalists and media houses are finding it difficult to be credible. Organizations and businesses are put under advanced fraud schemes. The platforms face the overwhelming moderation loads, and society, in general, suffers from the lack of trust in information and the democratic procedures. Such effects indicate the necessity of such a solution that is proactive, inclusive, and ethically accountable.

These issues are tackled by our proposed deepfake detector, which has a socially grounded design, and a technically innovative design. It has integrated an online detecting, multi-mode analysis, consent verification and forensic evidence creation into one system that is user-friendly and sensitive to privacy. The tool combines risk scores and explanation cues, which allows identifying the manipulated content, as well as, educating the user to recognize such content, creating digital resilience over the long-

term. This two-fold assurance of safety and enlightenment means that people are not only safeguarded but given the strength to explore the digital world with more self-confidence.

The short-term advantages of our product are extensive: victims will receive quick relief and access to legal assistance, journalists will protect their integrity, organizations will avoid fraud that costs a lot of money, platforms will have effective moderation filters, and the police will be able to obtain quality forensic evidence. In the long-term perspective, the detector will help to regain confidence in digital infrastructures, reinvent democratic rhetoric, and encouraging the responsible use of AI. Our product, with its provision of ethical protection, including local-first processing, consent verification, transparency, and others, shows that AI can be applied to safeguard human rights instead of harming them.

Finally, the lesson that can be learned is one of the most important in this project: the struggle against deepfakes is not a fight only against technical issues but also against moral, cultural, and political ones. It is not sufficient to detect but need solutions that are able to balance other conflicting values: free expression and harm reduction, speed and accuracy, privacy and transparency. We strike a very considerate balance between these, thereby making sure that not only do we mitigate the bad of today, but also that we set the example of responsible AI development in the future.

To sum up, our deepfake detector is not only a technical solution but also a moral intervention. It demonstrates how through the concepts of fairness, transparency and serving the greater good, innovation can transform AI into an agent of risk to be an agent of protection and empowerment. Our solution will help to create a healthier, safer, and more trustful digital society by addressing the harms of deepfakes at the core and backing the stakeholders who have been impacted most.

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